
TASKORA :AI-POWERED SMART TASK MANAGEMENT SYSTEM

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ABSTRACT

Taskora is an AI-powered smart task management system designed to automate task handling using intelligent agents. Unlike traditional systems that require manual task entry, Taskora automatically extracts tasks from user communications such as emails and Telegram messages using natural language processing techniques. The system prioritizes tasks, estimates effort, and provides smart reminders through a voice-enabled desktop assistant.

The system integrates a React-based frontend, FastAPI backend, SQLite database, and a standalone desktop agent. It uses machine learning models such as LSTM/BERT for task detection and prediction. Taskora also includes a self-learning mechanism that improves task estimation over time.

The solution enhances productivity by reducing manual effort, providing intelligent reminders, and enabling natural voice interaction.

Keywords: AI, Task Management, NLP, Voice Assistant, Machine Learning, Automation

INTRODUCTION

In today's fast-paced digital world, individuals constantly juggle multiple tasks from sources like emails, messages, and daily schedules, making efficient management a major challenge. Traditional task management systems rely heavily on manual input, requiring users to enter, organize, and track tasks themselves, which is time-consuming and often leads to missed deadlines, reduced productivity, and poor prioritization. With the advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP), there is a growing opportunity to create smarter systems that can automate task handling and understand user context. Taskora is an AI-powered smart task management system designed to address these issues by automatically detecting tasks from communications such as emails and Telegram messages, assigning priorities, and estimating effort using NLP techniques. It also includes a voice-enabled desktop assistant for natural interaction, along with a smart reminder system that notifies users at the right time without overwhelming them. Furthermore, Taskora incorporates self-learning capabilities by analyzing user behavior and refining predictions over time, making the system increasingly personalized and efficient. Overall, Taskora transforms traditional task management into an intelligent, automated, and user-centric experience that enhances productivity and reduces manual effort

Literature Review

2.1 AI in Task Management

- In recent years, Artificial Intelligence has significantly transformed productivity and task management systems. Traditional applications primarily focus on manual task entry and simple reminders, whereas AI-driven systems introduce automation, intelligence, and personalization. Technologies such as Natural Language Processing (NLP) enable systems to understand human language, extract meaningful information, and automate task-related activities.
- Machine learning models, including LSTM and transformer-based architectures like BERT, are widely used for text classification, intent detection, and context understanding. These models improve over time by learning from user behavior and historical data, making them suitable for applications such as task prioritization, recommendation systems, and predictive analytics.
- AI-based assistants also incorporate voice interaction, allowing users to communicate naturally with systems. Speech recognition and text-to-speech technologies enhance accessibility and usability, making task management more intuitive and efficient.

2.2 Existing Solutions

Several task management and productivity tools are currently available, but they have certain limitations

System	Key Features	Limitations
Traditional To-Do Apps	Manual task entry, basic reminders	No automation, no intelligence
Google Tasks	Simple interface, reminders	Limited AI features, no task prediction
Notion AI	Smart notes, AI suggestions	Not fully automated, requires manual input
Microsoft To Do	Task organization, syncing	No intelligent task detection
Voice Assistants	Voice commands	Not specialized for task automation

.Table 1: Comparison of existing knowledge management solutions.

Most of these systems require users to manually create and manage tasks. While some include AI features, they are not proactive and do not automatically detect tasks from communication sources.

2.3 Research Gap

Although existing systems provide basic task management and limited AI support, there is a lack of integrated solutions that combine automation, intelligence, and natural interaction. The major gaps identified are:

- Absence of automatic task extraction from communication platforms
- Lack of intelligent priority prediction and effort estimation
- Limited or no adaptive reminder systems
- Minimal integration of voice-based interaction in task management

No self-learning mechanisms to improve system performance over time.

Methodology

The Taskora system follows a structured methodology to enable intelligent and automated task management. It begins with data ingestion, where inputs are collected from user communication sources such as emails (via IMAP) and Telegram messages. These inputs are then cleaned and processed using Natural Language Processing (NLP) techniques to extract meaningful and actionable tasks. In the next stage, advanced AI models such as LSTM and BERT analyze the extracted data to understand context, classify tasks, assign priority levels (High, Medium, Low), and estimate the effort or time required for completion. This processed information is then stored in a structured format within an SQLite database, ensuring efficient organization and retrieval of task-related data.

Once the tasks are stored, a scheduling mechanism powered by APScheduler manages smart reminders by triggering notifications at appropriate times based on task priority and user behavior. A voice-enabled desktop assistant further enhances interaction by providing reminders, updates, and allowing users to control the system through speech. The system also incorporates a self-learning mechanism that continuously improves its performance by comparing predicted task durations with actual completion times and adapting future predictions accordingly. The overall architecture is modular, consisting of a React-based frontend, FastAPI backend, AI processing layer, database, and desktop agent, ensuring scalability, performance, accuracy, and a seamless user experience while maintaining data security and privacy.

CONTENT

Taskora is an AI-powered smart task management system designed to address the challenge of handling fragmented and unorganized tasks from multiple communication sources in today's digital environment. Users often receive tasks through emails, messages, and daily schedules, which are difficult to manage using traditional systems that rely on manual input and basic organization. Taskora overcomes these limitations by using advanced Natural Language Processing (NLP) techniques along with AI models such as LSTM and BERT to automatically extract tasks from sources like Email and Telegram, understand their context, and convert them into structured task entries. The system intelligently assigns priorities (High, Medium, Low) and estimates task duration, enabling users to focus on what matters most without manual effort.

Once tasks are processed, Taskora stores them in a structured database and uses a scheduling mechanism to trigger smart reminders based on priority and user behavior. A voice-enabled desktop assistant further enhances usability by allowing users to interact with the system through speech, receive updates, and manage tasks efficiently. The system follows a modular architecture

consisting of a React-based frontend, FastAPI backend, AI processing layer, database, and desktop agent, ensuring scalability and smooth performance. Additionally, Taskora incorporates a self-learning mechanism that improves over time by analyzing user activity and refining predictions. This approach results in a more personalized, efficient, and proactive task management experience, significantly improving productivity and reducing manual workload

Feature	Traditional Apps	Taskora
Task Entry	Manual	Automatic
Priority Setting	User-defined	AI-based
Reminders	Fixed	Smart & adaptive
Voice Interaction	Limited/None	Fully supported
Learning Capability	No	Yes

Table 1: Table Caption.

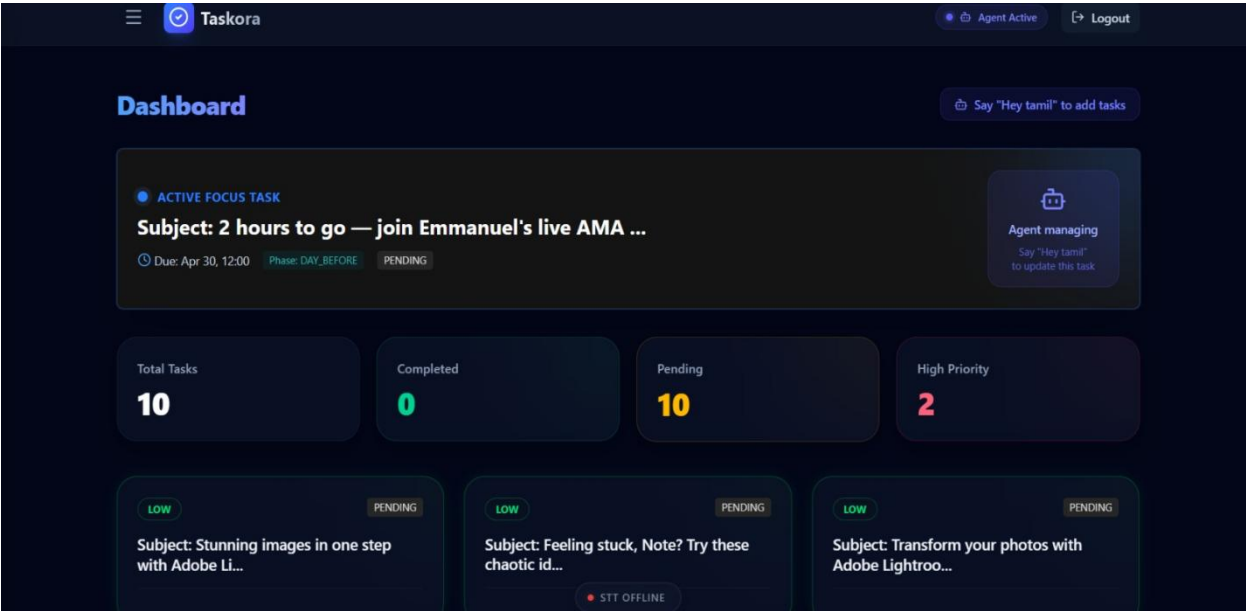


Fig. 1: Dashboard.

Results & Discussion

Taskora was tested across all major functionalities to ensure proper system behavior, and it successfully performs automatic task extraction from Email and Telegram, accurate priority assignment (High, Medium, Low), voice interaction through the desktop assistant, smart reminder notifications based on task importance, and real-time task updates in the dashboard, with all modules working smoothly and integrating effectively between the frontend, backend, and desktop agent. In terms of performance, the system demonstrates efficient real-time operation, with task extraction response times of approximately 2–3 seconds, high AI prediction accuracy for structured tasks, timely and reliable reminder scheduling, and a smooth, responsive user interface capable of handling moderate workloads without noticeable delays. Key achievements of the system include the successful development of an AI-based automated task management platform, implementation of a voice-controlled assistant with wake-word activation, intelligent task prioritization and effort estimation, a smart non-intrusive reminder system, and seamless integration of AI, UI, backend, and voice components. However, the system has certain limitations, such as support limited to Email and Telegram integrations, potential variability in voice recognition accuracy in noisy environments, dependency on system resources for running the desktop agent, and the absence of multi-user or collaborative features.

CONCLUSION

Taskora successfully demonstrates the application of Artificial Intelligence in automating task management. The system overcomes the limitations of traditional task management tools by eliminating manual task entry and introducing intelligent automation. Through the use of Natural Language Processing and machine learning models, Taskora can automatically extract tasks from user communications, assign priorities, and estimate effort effectively. The integration of a voice-enabled desktop assistant enhances user interaction by allowing natural communication with the system. The smart reminder mechanism ensures timely notifications without overwhelming the user, thereby improving productivity. Additionally, the self-learning capability enables the system to adapt and improve its predictions over time based on user behavior. Overall, Taskora functions as a proactive AI assistant that simplifies task management, reduces user effort, and provides an efficient and intelligent solution for handling daily activities.

Future Scope

- **Mobile Application:**
Develop Android and iOS applications for better accessibility
- **Advanced AI Models:**
Integrate more advanced models for improved accuracy in task detection and prediction
- **Multi-Platform Integration:**
Extend support to platforms like WhatsApp, Slack, and Google Calendar
- **Multi-User Collaboration:**
Enable shared task management and team collaboration features
- **Cloud Deployment:**
Deploy the system on cloud infrastructure for scalability and remote access
- **Offline Support:**
Implement offline functionality with synchronization when connected

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